Practice workbook · Using the cosine rule, including when it gives a quadratic

Using the cosine rule with two sides and the included angle to find the third side, or with three sides to find an angle, including in terms of x.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-07

1. Question 4 · part whole · 5 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.

4 In triangle ABC, $AB = 5x$ cm, $BC = (3x - 1)$ cm, $AC = (2x + 5)$ cm and angle $ABC = 60°$

Find, to 3 significant figures, the value of x.

(5)

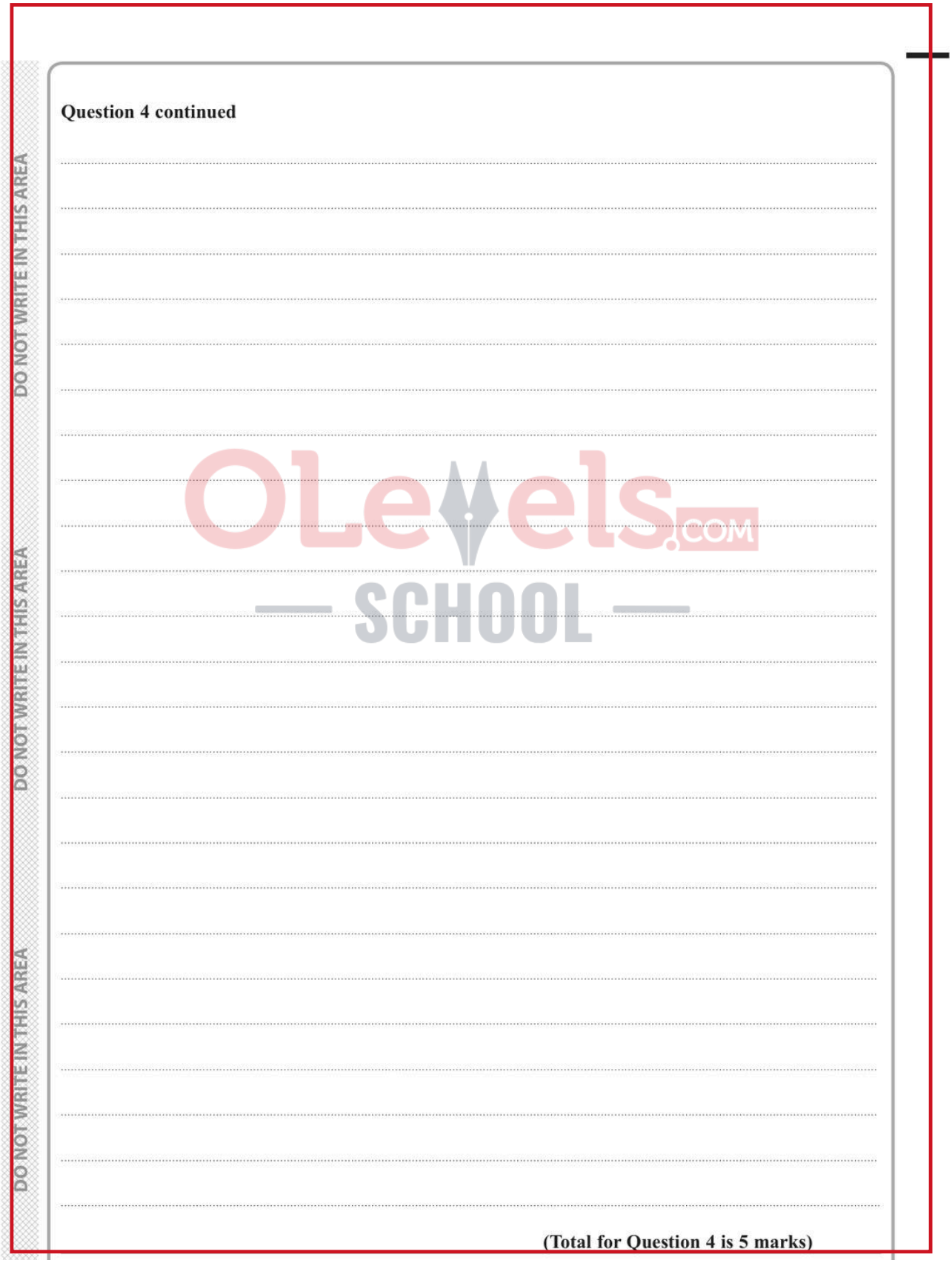
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1. Question 4 · part whole · 5 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.



Question 4 continued

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(Total for Question 4 is 5 marks)

1. Question 4 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 3 · part a · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

3

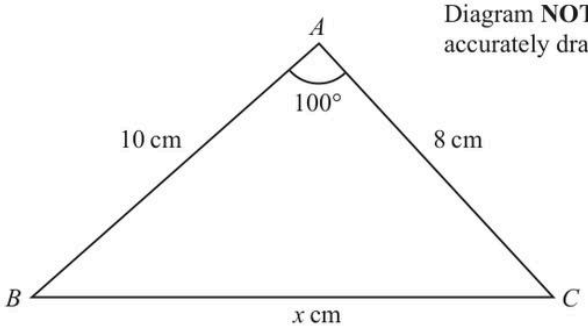


Figure 2

Figure 2 shows triangle ABC where

$$AB = 10 \text{ cm}, \quad AC = 8 \text{ cm}, \quad BC = x \text{ cm} \quad \text{and} \quad \angle BAC = 100^\circ$$

(a) Find, to 3 significant figures, the value of x

(2)

(b) Find, in degrees to one decimal place, the size of

(i) angle ABC (ii) angle ACB

(3)

The bisector of angle ABC meets AC at the point M

(c) Find the area, in cm^2 to 3 significant figures, of triangle BMC.

(4)

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P 7 4 0 9 8 A 0 6 3 2

2. Question 3 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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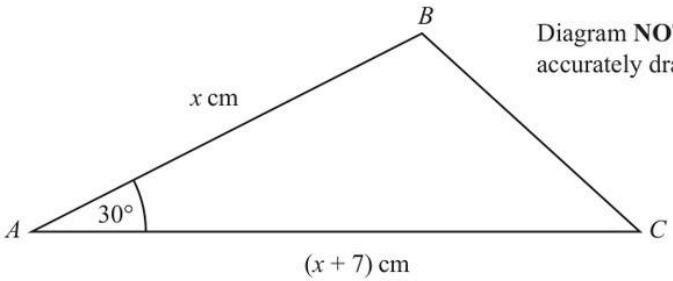


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$

(3)

(b) Find, in cm to 3 significant figures, the length of BC

(2)

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$

(3)

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P 7 4 0 9 9 A 0 6 3 2

3. Question 3 · part b · 2 marks · Working and answer

Working and answer

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4. Question 8 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

8 The point A has coordinates (2, 6), the point B has coordinates (6, 8) and the point C has coordinates (4, 2).

(a) Find the exact length of

(i) AB (ii) BC (iii) AC

(4)

(b) Find the size of each angle of triangle ABC in degrees.

(3)

The points A, B and C lie on a circle with centre P.

(c) Find the coordinates of P.

(2)

(d) Find the exact length of the radius of the circle in the form $\sqrt{a}$, where a is an integer.

(2)

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4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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5. Question 3 · part a · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

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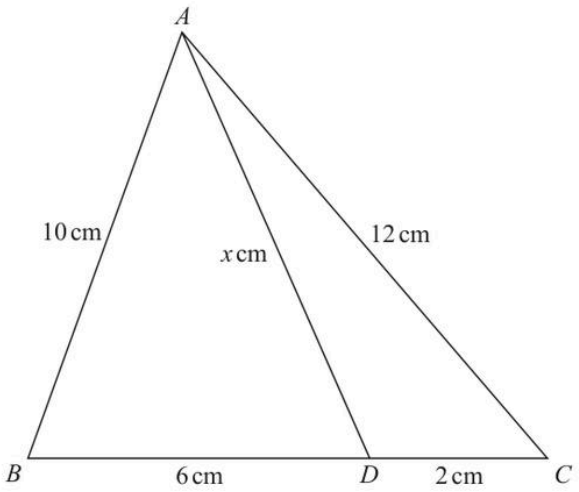


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the triangle ABC in which $AB = 10\,\text{cm}$ and $AC = 12\,\text{cm}$.
The point D lies on BC such that $BD = 6\,\text{cm}$, $DC = 2\,\text{cm}$ and $AD = x\,\text{cm}$.

(a) Show that $x = 11$

(4)

(b) Find the area, in cm^2 to 3 significant figures, of triangle ADB.

(4)

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5. Question 3 · part a · 4 marks · Working and answer

Working and answer

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6. Question 9 · part a · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

9

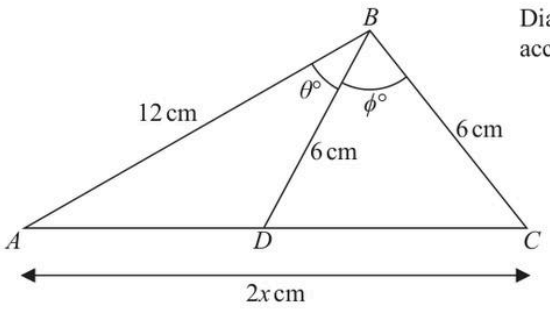


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Figure 3

Figure 3 shows triangle ABC with $AB = 12\,\text{cm}$, $BC = 6\,\text{cm}$ and $AC = 2x\,\text{cm}$.

The point D is the midpoint of AC and $BD = 6\,\text{cm}$.

$\angle ABD = \theta°$ and $\angle DBC = \phi°$ where $\theta \neq 0$ and $\phi \neq 0$

(a) Show that $\cos ADB = \dfrac{x^2 - 108}{12x}$

(2)

(b) Hence, or otherwise, show that $AC = 6\sqrt{6}\,\text{cm}$.

(4)

(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$

(4)

(d) Hence show that $\theta = 180 - 2\phi$

(2)

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6. Question 9 · part a · 2 marks · Working and answer

Working and answer

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7. Question 7 · part a · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

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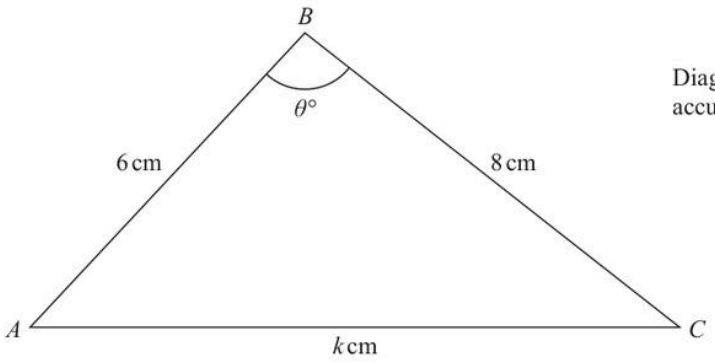


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC

$AB = 6\,\text{cm}$ $BC = 8\,\text{cm}$ $AC = k\,\text{cm}$ $\angle ABC = \theta°$

(a) Show that $\cos\theta° = \dfrac{100 - k^2}{96}$ (2)

The area of triangle ABC is $\sqrt{455}$ cm^2

(b) Find the two possible values of k (7)

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7. Question 7 · part a · 2 marks · Working and answer

Working and answer

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8. Question 2 · part a · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

2 In triangle ABC, $AB = 3x$ cm, $BC = 5x$ cm and $\angle ABC = 110°$

(a) Find, in degrees to one decimal place, the size of $\angle BCA$

(4)

The area of triangle ABC is 24 cm^2

(b) Find, to 3 significant figures, the value of x

(3)

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8. Question 2 · part a · 4 marks · Working and answer

Working and answer

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9. Question 2 · part whole · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

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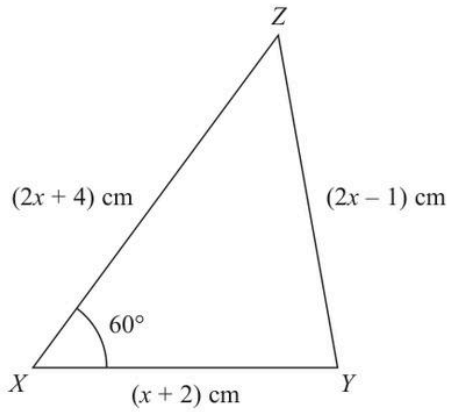


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle XYZ in which

$XY = (x + 2)$ cm $\quad XZ = (2x + 4)$ cm $\quad YZ = (2x - 1)$ cm $\quad$ and $\angle YXZ = 60°$

Find the value of x

Give your answer in the form $p + q\sqrt{3}$ where p and q are integers to be found.

(4)

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9. Question 2 · part whole · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 4 marks)

9. Question 2 · part whole · 4 marks · Working and answer

Working and answer

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10. Question 5 · part a · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

5 In triangle ABC, $AC = 12$ cm, $BC = 14$ cm, $AB = x$ cm and angle $ABC = 30°$

(a) Show that $x = P\sqrt{3} \pm \sqrt{95}$ where P is a prime number.

(5)

(b) Hence or otherwise, find in cm^2, the difference between the two possible areas of triangle ABC

Give your answer in the form $m\sqrt{n}$ where m is a prime number and n is an integer.

(3)

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10. Question 5 · part a · 5 marks · Working and answer

Working and answer

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11. Question 3 · part a · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

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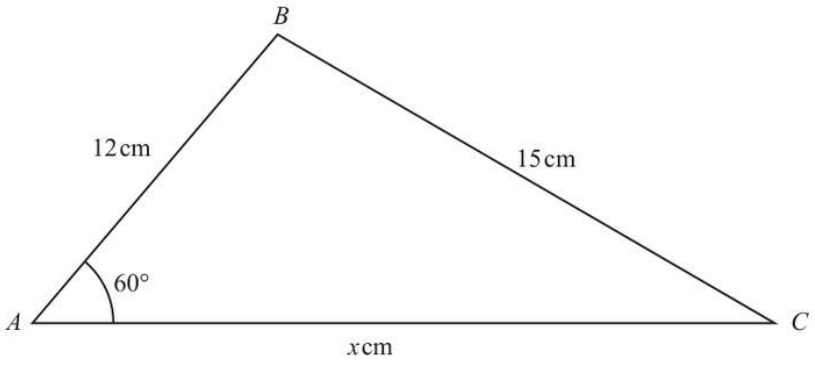


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Figure 1

Figure 1 shows triangle ABC where

$AB = 12$ cm $\quad BC = 15$ cm $\quad AC = x$ cm $\quad \angle BAC = 60°$

(a) Show that x satisfies the equation

$$x^2 - 12x - 81 = 0$$

(3)

(b) Hence find the exact area, in cm^2, of triangle ABC

Give your answer in the form $9\left(p\sqrt{q} + \sqrt{r}\right)$ where p, q and r are integers.

(3)

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11. Question 3 · part a · 3 marks · Working and answer

Working and answer

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12. Question 9 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

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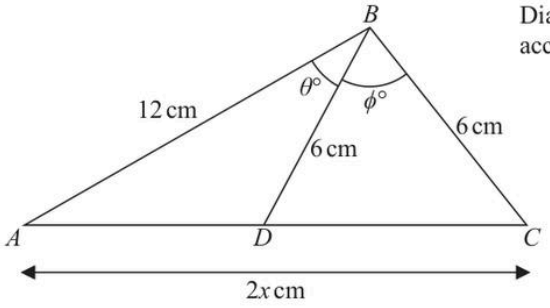


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle ABC with $AB = 12\,\text{cm}$, $BC = 6\,\text{cm}$ and $AC = 2x\,\text{cm}$.

The point D is the midpoint of AC and $BD = 6\,\text{cm}$.

$\angle ABD = \theta°$ and $\angle DBC = \phi°$ where $\theta \neq 0$ and $\phi \neq 0$

(a) Show that $\cos ADB = \dfrac{x^2 - 108}{12x}$

(2)

(b) Hence, or otherwise, show that $AC = 6\sqrt{6}\,\text{cm}$.

(4)

(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$

(4)

(d) Hence show that $\theta = 180 - 2\phi$

(2)

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12. Question 9 · part b · 4 marks · Working and answer

Working and answer

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